

instrument meteorological conditions (IMC)

In aviation, an official category of poor weather conditions in which a pilot will necessarily have to fly mainly by reference to cockpit instrumentation, rather than by direct visual means.

instrument rating

A qualification that permits a pilot to fly in situations where flight depends on observing cockpit instruments, rather than on visual cues.

interrupter gear

A system developed in World War I that allowed a fixed machine gun carried in an aircraft to fire through a propeller, by halting the gun's fire whenever a blade was in line with the barrel of the gun.

jet-assisted takeoff (JATO)

Using attached rockets to assist the takeoff of aircraft. The method was used for heavy aircraft and military gliders during World War II and also in the early days of jet airplanes, but is less common today.

jet efflux

The discharge of exhaust gases from a jet, especially in the context of possible hazardous effects.

jet engine

See turbojet.

joystick

A hand-operated lever in an airplane's cockpit that controls the ailerons and elevator. (The rudder is controlled separately by foot pedals.) *See also* ailerons, elevator.

king post

A vertical post fixed on top of the fuselage of some older aircraft, from which wires are stretched to help support the wings. It is also used in some hang gliders.

landing gear

The arrangement of wheels, skids, or floats on which an aircraft touches down when landing. The landing gear arrangements of wheeled airplanes can be divided into those that are taildraggers and those that have a tricycle gear. *See also* taildraggers, tricycle gear.

leading edge

The front edge of a wing that cuts through the air first.

lift

The force exerted on a moving airfoil that causes a wing to rise.

liquid cooling

A method of cooling an engine via a system of circulating liquids. It is used when cooling by using air alone would not be adequate.

Mach number

The ratio of an aircraft's speed to the speed of sound. Mach 2, for example, is twice the speed of sound.

magneto

A device that provides high-voltage electricity for engine ignition using rotating permanent magnets, without the need for a separate battery or electricity supply. Although superseded in automobiles, it is still commonly used in aircraft engines because of its reliability.

Medevac

Short for medical evacuation. The term came into use during the Vietnam War, and was used to mean both the evacuation of casualties from a battle zone and a helicopter used for this purpose.

monocoque

A term referring to a method of constructing an aircraft's airframe (particularly its fuselage) so that the main strength is provided by the aircraft's outer shell rather than by internal struts or beams.

monoplane

An airplane with only a single pair of wings. *See also* biplane.

nacelle

A streamlined casing that supports and surrounds an engine or other structure.

nose wheel

A single landing wheel positioned beneath the nose section of an airplane's fuselage. *See also* tricycle gear.

oleo strut

A vertical strut often connecting the nose wheel to the fuselage that contains hydraulic fluid and compressed air and acts as a shock absorber on landing.

ornithopter

A flying machine created with wings that flap, in imitation of bird flight. Although many pre-20th-century ideas for heavier-than-air flying machines were for ornithopters, they have never been proved to work in practice.

overhead camshaft (OHC)

A camshaft positioned at the top of a piston engine's cylinder block, close to the valves, which it can control directly without an additional system of pushrods. *See also* camshaft, pushrod.

overhead valves (OHV)

Valves on the cylinder head of a piston engine that are operated by pushrods actuated by a camshaft situated below the cylinder head.

parasol wing

An arrangement in which a single continuous wing is positioned above the top of an aircraft's fuselage and connected to it by struts. One disadvantage is that the struts cause increased drag.

payload

The load carried by an aircraft, including passengers and cargo, from which revenue is obtained.

piston engine

An engine in which the fuel ignition takes place in cylindrical combustion chambers (cylinders), each furnished with a close-fitting movable piston. After ignition, the hot expanding gases push up the piston, and its up-and-down motion is converted by a piston rod and crankshaft into rotary motion, which is used to drive a propeller. An aircraft piston engine always has several cylinders that fire in turn to create a continuous rotary force.

pitch

The vertical movement of an aircraft's nose in relation to its tail. It is controlled by the aircraft's elevators.

pitot tube

A small, open-ended tube, usually located on the leading edge of the wing, that is used to measure the aircraft's speed and is linked to a pressure-measuring device in the cockpit.

powerplant

The permanent assembly—including the propellers and engine—that is responsible for aircraft propulsion.

primary flight display (PFD)

The central display screen in the cockpit of modern aircraft, showing the data from the most important aircraft instruments on a single screen.

propeller

In aircraft, a rotating device consisting of shaped blades that is designed to accelerate air backward and thus pull the aircraft forward. Blades set at different angles (pitches) work best at different air speeds, and so many propellers have variable pitch control to allow this. Constant speed propellers are able to produce different power outputs as required, while keeping their own rotational speed constant. The rotation of propellers can cause problems of twisting forces being applied to the aircraft or its engines. These can be countered by contra-rotating propellers (two propellers one behind the other on the same engine, rotating in opposite directions) or counter-rotating propellers (propellers on opposite wings rotating in opposite directions).

pursuit aircraft

An older name for a fighter aircraft in use mainly in the United States before World War II.

pushrod

A rod used to transmit movement from a camshaft to the valves in some piston engines. *See also* overhead valves.

radome

A protective radar dome that allows radio waves in and out.

ramjet

A jet engine that is essentially an open pipe. It does not require a gas turbine, but relies on the rush of incoming air itself to create sufficient compression for combustion. It cannot operate at all at slow speeds and works best at speeds of Mach 3 to Mach 6. For this reason its main use is as a supplementary engine in some supersonic military and experimental craft.

ramp doors

The doors inside the intake ramps of a jet aircraft's engine. These regulate the flow of air to the engine.

reduction gear

Gears arranged so that high-speed rotary motion is converted into slower but more powerful motion. A reduction gear is essential in turboprop engines where the propellers are required to turn at a slower speed than the engine's turbine wheels. *See also* turboprop.

roll

The rotating motion of an aircraft's fuselage in which one wing tends to rise as the other one dips. It is controlled by the aircraft's ailerons. *See also* ailerons.

rotary engine

A type of aircraft piston engine in which the engine cylinders are arranged radially and rotate along with the propeller, cooling themselves in the process.

rotor

A rotating device. A term used to describe the assembly of blades that lifts a helicopter, and the rotating wheel(s) of a turbine. *See also* stator.

rotor craft

Any aircraft that gains its lift via a system of rotating airfoil blades. Most rotor craft are helicopters, but the term also includes autogyros and some other aircraft. *See also* autogyro.

rpm

Stands for revolutions per minute, the measure of the rotation speed of a propeller, shaft, or engine.

rudder

A control surface usually situated on an airplane's vertical tail fin. Its main functions are to control yaw and to help turn the aircraft.

sailplane

Another name for a glider, especially one with rigid wings that is used for recreation and is capable of soaring in updrafts of air, as distinct from the heavier gliders formerly used for military purposes.

Schneider Trophy

A trophy awarded, together with a cash prize, for the winner of what became an annual series of seaplane races sponsored by wealthy Frenchman Jacques Schneider. Races were held between 1913 and 1931 and did much to encourage the development of higher standards in aircraft design.

scout

Originally a term used to describe a reconnaissance plane, but later also applied to a lightly armed fighter plane.

service ceiling

The maximum altitude at which an aircraft's rate of climb falls below 100ft per minute.

servo tab

A controllable, leverlike structure attached to some aircraft control surfaces. It moves in the opposite direction to the control surface itself, and has the effect of decreasing the amount of force needed to move the control surface. *See also* anti-servo tab, control surface.

sesquiplane

A biplane in which one pair of wings is much smaller than the other. *See also* biplane.

slats (leading-edge)

Movable control surfaces on the leading edge of an airplane's wings. They function by increasing lift at lower air speeds, especially when the plane is landing, by allowing the aircraft to fly at a higher angle of attack without stalling. *See also* angle of attack, stall.

smart weapon

A precision-guided munition that is directed to its target using laser guidance or, more recently, satellite-linked GPS (global positioning system) guidance.